

Course: Data Science Laboratory**Course Code: 303105394****Prerequisite:** Basic Knowledge of Programming

Rationale : This course provides necessary knowledge on data manipulation and to perform analysis on the practical problems using statistical and machine learning approach to generate report and visualize the results in graphical form using programming tool.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, T - Theory, P - Practical

List of Practical	
1.	House rent prediction using linear regression.
2.	Medical diagnosis for disease spread pattern.
3.	Automate email classification and response.
4.	Customer segmentation in business model based on their demographic, psychographic and behavior data.
5.	Construct a recommendation system based on the customer transaction using Association rule mining.
6.	Behavioral analysis of customers for any online purchase model.
7.	Agricultural data analysis for yield prediction and crop selection on Indian terrain data set.
8.	Develop a business model to predict the trend in Investment and Funding.

Course Outcome
After Learning the Course the students shall be able to:
1. Ability to gain basic knowledge on data science.
2. Convert the real time data into suitable form for analysis.
3. Gain the insights from the data through statistical inferences.
4. Develop suitable models using machine learning techniques and to analyze its performance.
5. Identify the requirement and visualize the results.
6. Analyze on the performance of the model and the quality of the results.